

Contrastive Mixture Diffusion Models

Jen-Tzung Chien , Senior Member, IEEE, and Chih-Chun Chen

Abstract—Continuous diffusion model is feasible to perform smoothed transitions for flexible text generation in a continuous latent space. A challenging issue in continuous diffusion is to implement the denoising process and manipulate the generated tokens to reflect physical phenomenon in diffusion steps towards a contextually and semantically meaningful sentence. Basically, a high-quality text generation can be fulfilled by an easy-first strategy, which prioritizes the generation of high-frequency simple or general tokens in the beginning and then low-frequency complex or specific tokens in the end via mask language model. This paper introduces the mask noise as an absorbing state and develops a new continuous-discrete denoising process in a diffusion model. Each diffusion step is performed by generating either a continuous word embedding or a discrete mask token where the latent transition is modeled by a Gaussian-Dirac mixture distribution. The easy-first text generation is then implemented and strengthened via a contrastive loss to disentangle the generation between simple and complex tokens. A contrastive mixture diffusion model is accordingly exploited by minimizing a variational bound of negative log likelihood, which is regularized by aligning the denoising network posterior with the Gaussian-Dirac posterior in each diffusion transition based on an approximate Kullback-Leibler divergence. The experiments on text paraphrasing and other tasks demonstrate the effectiveness and efficiency of sentence generation by using the proposed method where the easy-first strategy in generation behavior is illustrated.

Index Terms—Contrastive diffusion, continuous-discrete denoising process, easy-first generation, Gaussian-Dirac mixture distribution.

I. INTRODUCTION

TRANSFORMERS have been the primary tools for natural language processing tasks. Recently, diffusion models [1] have emerged as a promising alternative for text generation. The use of discrete diffusion for text generation was first introduced in [2] where the discrete state space following Markov transition was developed. Some other alternatives [3], [4] have taken a different approach to draw tokens where continuous diffusion within a continuous embedding space was performed and implemented by the advanced sampling methods [5], [6], [7], enabling efficient generation, while discrete diffusion models were trained with the ability to represent the inherent discrete structure of the language [2], [8]. A notable example was called the sequence-to-sequence diffusion model [9], which integrated

both continuous and discrete noise types and employed a fast solver [7] to accelerate the hybrid continuous and discrete diffusion process.

Recent studies have shown that the model performance can be improved by focusing on generating simpler or more common tokens earlier in the reverse diffusion process. This approach towards the easy-first tendency aligns the generated text more closely with the input condition, resulting in better fluency and coherence [8], [10]. In [10], the noise level was adjusted based on the token entropy while the score driven by term frequency and inverse document frequency was utilized to prioritize certain tokens in the generation process. The noise schedule was crafted to simulate a human to generate the content, beginning with simpler elements and progressively refining the output.

This paper presents a new continuous-discrete denoising process in a continuous diffusion model where the mask tokens are arranged as the discrete noise signals and the mask language model is implemented in a denoising transition based on a Gaussian-Dirac mixture distribution [11]. This framework is carried out by mimicking an easy-first generation through contrastive learning [12] over those simple and complex tokens with high and low occurrence frequencies, respectively. Such a strategy is consolidated to prioritize text generation to draw coherent and natural paraphrases. A contrastive mixture diffusion model is constructed.

In addition, this study addresses a key challenge regarding the treatment of hybrid noise in diffusion models due to the nature of mask tokens [13], [14], [15], [16] accompanied with noisy word embeddings. While the previous models primarily assumed either the continuous loss function based on the Gaussian noises or the discrete loss function based on the uniform noises, the hybrid nature of continuous-discrete diffusion model substantially complicates the derivation of denoising process for text generation. This paper introduces a new approximation to an alignment loss as a Kullback-Leibler divergence between forward and reverse processes at each diffusion step that accounts for both Gaussian and Dirac mixture components in the hybrid noise setting, providing a meaningful approach to reflect and optimize the model. Such a treatment ensures that the hybrid diffusion is feasible to handle both continuous and discrete noises effectively, leading to an improved text generation.

This paper addresses several limitations in previous approaches and offers a direction to improve quality and diversity in the generated tokens. A number of findings are explored and summarized as follows.

- a new contrastive mixture diffusion model is presented by integrating continuous and discrete noises as the absorbing noise to build large language model.

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The authors are with the Institute of Electrical and Computer Engineering, National Yang Ming Chiao Tung University, Hsinchu City 30010, Taiwan (e-mail: jtchien@nycu.edu.tw).

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- a Gaussian-Dirac mixture distribution is proposed to characterize both continuous and discrete noises in a diffusion model for mask language model.
- an alignment loss between forward and reverse processes at each diffusion step based on an approximate Kullback-Leibler divergence is derived.
- an easy-first generation is fulfilled through a contrastive diffusion model where a triplet loss for high-frequency and low-frequency tokens relative to anchor mask in embedding space is minimized.
- a set of experiments on text generation is evaluated to examine the continuous-discrete denoising process in terms of sentence similarity and semantic quality, and the generation behavior in an easy-first fashion.

The remaining of this paper is organized as follows. Section II addresses the basics of diffusion process from continuous, discrete to continuous-discrete variants. Section III presents the construction of Gaussian-Dirac diffusion and its implementation in denoising process, optimization problem and training algorithm. Section IV reports the evaluation results and ablation studies on text generation. Finally, the conclusions drawn from this study are given in Section V.

II. PROBABILISTIC DIFFUSION MODELS

Text generation using the generative pre-trained transformer (GPT) is operated as an autoregressive machine. Diffusion model is attractive as a generative sequence model, feasible to generate the whole sequence in a non-autoregressive manner, and has achieved remarkable performance in image, speech and text generation [17], [18], [19], [20].

A. Continuous Diffusion Model

Generative model aims to generate or draw data sample or string $\mathbf{x}$ from an underlying distribution $p(\mathbf{x})$, which is constructed and estimated by solving a latent variable model through the approximate inference [21], [22], [23]. The variational autoencoder (VAE) [21] was developed to reconstruct an observation $\mathbf{x}$ from a Gaussian posterior $p_\theta(\mathbf{x}|\mathbf{z})$ given by a latent variable $\mathbf{z}$ with the parameter θ calculated from a decoder. The latent variable $\mathbf{z}$ is drawn from variational distribution $q_\phi(\mathbf{z}|\mathbf{x})$ with parameter ϕ , which is determined via an encoder by feeding original sample $\mathbf{x}$. Encoder ϕ and decoder θ are trained by maximizing the evidence lower bound of log likelihood from data collection $\mathcal{D} = \{\mathbf{x}\}$ [24].

Diffusion model also runs an encoding (or forward) process and a decoding (or reverse) process, and is viewed as a multistep VAE consisting of a noising process and a denoising process. The forward process recursively adds noise to diffuse from a clean observation $\mathbf{x}_0$ following a Markov chain for generating the noisy latent variables by

$$q(\mathbf{x}_{1:T} \triangleq \{\mathbf{x}_t\}_{t=1}^T | \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1}) \quad (1)$$

where $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$ is given with a crafted variance schedule $\{\beta_t\}_{t=1}^T$. Such a noise process

preserves the variance and implements the noise addition

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \boldsymbol{\epsilon}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \alpha_t \triangleq 1 - \beta_t. \quad (2)$$

This transition process doesn't involve any learnable parameter. The latent sample at diffusion step t is then drawn by

$$\mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad (3)$$

where $\bar{\alpha}_t \triangleq \prod_{i=1}^t \alpha_i$. After adding sufficient times of noise $T \rightarrow \infty$, the distribution $q(\mathbf{x}_T | \mathbf{x}_0)$ of noisy latent variable $\mathbf{x}_T$ eventually becomes a standard Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$ as the stable distribution. In forward pass, β_t controls the amount of adding noise with a noisy schedule, which has a small value to ensure the reverse transition to be Gaussian [25].

The reverse process is seen as the reverse direction of forward process. This is a sampling process, which initially draws a standard Gaussian sample $\mathbf{x}_T \sim p(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and gradually denoises the noisy latent variable $\mathbf{x}_t$ via a series of diffusion steps back to its original observation $\mathbf{x}_0$. Relative to the noising process in (1), the Markov transition from a noise sample $\mathbf{x}_T$ towards reconstructing a clean sample $\mathbf{x}_0$ through denoising process is yielded by

$$p(\mathbf{x}_{0:T-1} | \mathbf{x}_T) = \prod_{t=1}^T p(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (4)$$

However, the true reverse transition distribution $p(\mathbf{x}_{t-1} | \mathbf{x}_t)$ is unknown. This distribution is constructed by considering the forward transition distribution in reverse direction as

$$\begin{aligned} q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) &\propto q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0) q(\mathbf{x}_{t-1} | \mathbf{x}_0) \\ &= \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\boldsymbol{\beta}}_t \mathbf{I}). \end{aligned} \quad (5)$$

In (5), the derivation is done for $\tilde{\beta}_t \triangleq \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$ and

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t}. \quad (6)$$

Accordingly, the reverse transition distribution for sampling with denoising parameter θ is approximated as

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \tilde{\boldsymbol{\beta}}_t \mathbf{I}). \quad (7)$$

The Gaussian mean $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)$ is obtained by treating the output of a neural network denoiser $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$ as the estimated clean sample $\hat{\mathbf{x}}_0$ and substituting it into (6) to find $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) \leftarrow \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \hat{\mathbf{x}}_\theta)$, which is updated by feeding current sample $\mathbf{x}_t$ at diffusion step t . This denoising process, initialized from a white noise $\mathbf{x}_T$ and diffused towards an original sample $\mathbf{x}_0$, is recursively calculated by

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + (1 - \alpha_t)\boldsymbol{\epsilon}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (8)$$

A denoising neural network $\hat{\mathbf{x}}_0 \leftarrow \hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$ needs to be trained to fulfill a continuous diffusion model.

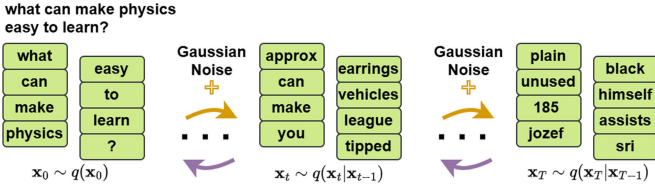


Fig. 1. Illustration of continuous diffusion process for a text string in latent space where Gaussian noise is added.

The denoising network f_θ or $\hat{x}_\theta$ is learned by minimizing a variational bound of negative log likelihood, derived by

$$\begin{aligned}
 -\log p_\theta(\mathbf{x}_0) &\leq \mathbb{E}_q \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] \\
 &= \underbrace{-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]}_{\mathcal{L}_{\text{rc}}} + \underbrace{\mathcal{D}_{\text{kl}}(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T))}_{\mathcal{L}_T} \\
 &\quad + \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\mathcal{D}_{\text{kl}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\mathcal{L}_{t-1}} \triangleq \mathcal{L}_{\text{diff}}
 \end{aligned} \tag{9}$$

where $\mathcal{D}_{\text{kl}}(q||p)$ is the Kullback-Leibler (KL) divergence between q and p , and $\{\mathbf{x}_t\}_{t=1}^T$ are latent variables of clean $\mathbf{x}_0$. In the derived diffusion objective $\mathcal{L}_{\text{diff}}$, the first term is the reconstruction loss $\mathcal{L}_{\text{rc}}$, which can be minimized by Monte Carlo method [21], the second term $\mathcal{L}_T$ is a constant, and the third term $\mathcal{L}_{\text{kl}}$ with $\mathcal{L}_{t-1}$ at each diffusion step $t-1$ is a matching term in denoising process, which is minimized to align between forward posterior $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ in (5) and (6) and reverse distribution $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ in (7).

Importantly, minimizing KL loss $\mathcal{L}_{t-1}$ corresponds to fulfill a learning strategy for a denoising network f_θ , which guides the denoised latent variables from $\mathbf{x}_t$ to $\mathbf{x}_{t-1}$ at each t to align with surrogate samples $\{\mathbf{x}_t\}_{t=0}^T$, which have been elegantly deteriorated from clean sample $\mathbf{x}_0$ towards white noise $\mathbf{x}_T$ in forward diffusion with sufficiently large T based on a predefined additive noisy transformation with variance schedule $\{\beta_t\}_{t=1}^T$. In general, KL divergence between two Gaussians $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ and $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ in $\mathcal{L}_{t-1}$ can be easily calculated in a form of Euclidean distance between their mean vectors $\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0)$ and $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)$, which were developed for continuous-valued image generation [26].

B. Continuous-Discrete Diffusion Models

Text sequence generation has been tackled by continuous and discrete diffusion models. In [6], [27], a diffusion language model (diffusion-LM) was proposed to generate text data where a continuous diffusion was run over those real-valued word embeddings in a text string. A clamping method was employed in sampling process by clamping or rounding the generated embeddings $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$ to their nearest word embeddings. This clamping scheme inevitably caused the rounding errors which limited the generation performance. Fig. 1 illustrates a continuous diffusion process for sequence generation with Gaussian noise.

The discrete diffusion model [2], called D3PM, was proposed to deal with natural language generation where the forward and reverse processes were characterized by transition probabilities $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ and $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ based on the discrete tokens $\mathbf{x}_t$ and $\mathbf{x}_{t-1}$ in one-hot vectors with K categories, modeled by the categorical distributions $\text{Cat}(\mathbf{x}_t; \mathbf{p} = \mathbf{M}_t \mathbf{x}_{t-1})$ and $\text{Cat}(\mathbf{x}_{t-1}; \mathbf{p} = \mathbf{p}_\theta(\mathbf{x}_t, t))$, respectively [20]. Here, $\mathbf{M}_t$ denotes the transition matrix with an entry $[\mathbf{M}_t]_{ij}$ showing a transition $q(\mathbf{x}_t = i|\mathbf{x}_{t-1} = j)$ of a word in the string from categories j to i and θ in $\mathbf{p}_\theta(\mathbf{x}_t, t)$ denotes the posterior categorical parameter with a denoising network f_θ by feeding a text string $\mathbf{x}_t$ to generate the enhanced string $\mathbf{x}_{t-1}$ at each diffusion step t . The transition matrix for noising process, given by $\mathbf{M}_t = (1 - \beta_t)\mathbf{I} + (\beta_t/K)\mathbf{1}\mathbf{1}^\top$ with a hyperparameter schedule $\beta_t \in [0, 1]$, is used to diffuse the original token $\mathbf{x}_0$ towards a pure noise $\mathbf{x}_T$ as a uniform distribution $\mathcal{U}$ with a sufficiently large T . $\mathbf{1}$ denotes a vector of all entries being 1. Similar to (5), (6), the forward transition in reverse direction is obtained by

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat} \left(\mathbf{x}_{t-1}; \mathbf{p} = \frac{\mathbf{M}_t^\top \mathbf{x}_t \odot \overline{\mathbf{M}}_{t-1} \mathbf{x}_0}{\mathbf{x}_t^\top \overline{\mathbf{M}}_t \mathbf{x}_0} \right) \tag{10}$$

where $\overline{\mathbf{M}}_t \triangleq \prod_{i=1}^t \mathbf{M}_i$. Diffusion objective $\mathcal{L}_{\text{diff}}$ in (9) is then minimized to estimate the denoising network f_θ .

An alternative solution to text generation was extended via sequence-to-sequence (Seq2Seq) learning [28], [29] based on the DiffuSeq [3] and DiffuSeq-v2 [9]. DiffuSeq was built by training a continuous diffusion model based on the concatenated embedding $\mathbf{z}_0 = g_\phi(\mathbf{x}_0 \oplus \mathbf{y}_0)$ with parameter ϕ of embedding encoder g_ϕ from a pair of condition sequence $\mathbf{x}_0$ and target sequence $\mathbf{y}_0$. DiffuSeq-v2 further implemented a diffusion transition for text generation by integrating both continuous and discrete noises based on the concatenated embeddings $\mathbf{z}_t$ from $\mathbf{x}_t$ and $\mathbf{y}_t$ where a paraphrasing task was evaluated. A soft mask token was introduced to address the mismatch between training and inference phases existing in commonly-used clamping trick [27] in Diffusion-LM. A Gaussian noise was used in the continuous diffusion via $q(\mathbf{z}_t|\mathbf{z}_0)$ by referring to (2), (3) to yield

$$\mathbf{z}_t = \sqrt{\alpha_t} \mathbf{z}_0 + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{11}$$

given by the original data embedding $\mathbf{z}_0$. Each token in $\mathbf{z}_t$ is randomly replaced during training by a learnable absorbing noise $\mathbf{m}$, acting as a mask token that has no probability to transit to other states in the noising process. The n -th token embedding of a string $\mathbf{z}_t$ becomes a latent mixture variable

$$\hat{\mathbf{z}}_t^n = \begin{cases} \mathbf{m}, & \text{if } \rho_t^n = 1 \\ \mathbf{z}_t^n, & \text{if } \rho_t^n = 0 \end{cases} \tag{12}$$

where $\rho_t^n \sim \text{Bernoulli}(\gamma_t)$ is sampled as a discrete mask indicator at step t and γ_t controls the mask ratio. DiffuSeq-v2 is seen as a continuous-discrete diffusion model where the mask token was merged in a continuous diffusion.

Notably, the KL loss $\mathcal{L}_{t-1}$ between $q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{z}_0)$ and $p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t)$, which are no longer Gaussians, does exist in DiffuSeq-v2 but was ignored and over-simplified in [9]. To clarify this influential issue, this paper presents a new continuous-discrete denoising process to provide a theoretical justification

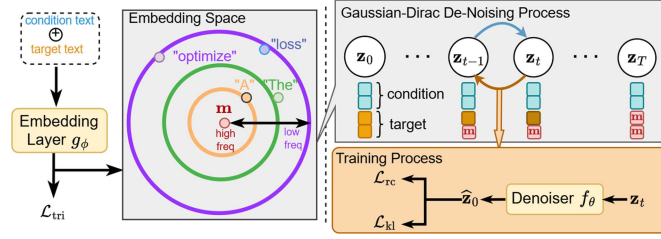


Fig. 2. Overview of calculating the learning objectives $\mathcal{L}_{tri}$, $\mathcal{L}_{rc}$ and $\mathcal{L}_{kl}$ for contrastive mixture diffusion where mask token $\mathbf{m}$, embedding encoder g_ϕ , and diffusion denoiser f_θ are trained. Forward process is to recursively impose an Gaussian-Dirac mixture noise to the embedding of target $\mathbf{y}_t$ through an embedding encoder g_ϕ at each step t while that of condition $\mathbf{x}_t$ remains unaltered. The target embedding in $\mathbf{z}_T$ for individual tokens eventually regresses towards mask token $\mathbf{m}$. Reverse process is to denoise current embedding $\mathbf{z}_t$, a concatenation of condition $\mathbf{x}_t$ and target embeddings $\mathbf{y}_t$, to clean embedding $\hat{\mathbf{z}}_0$ via a diffusion denoiser f_θ .

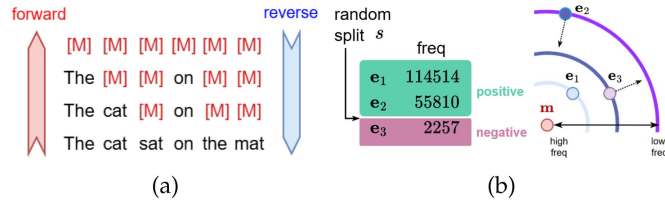


Fig. 3. (a) Illustration of an easy-first text generation via diffusion process between a clean sentence string and a noisy mask string of $\mathbf{m}$ or $[\mathbf{M}]$. (b) Implementation of a random split s to choose the paired positive and negative samples for contrastive learning. Concentric circles imply how the embeddings $\mathbf{e}_2$ and $\mathbf{e}_3$ with higher and lower word frequencies are updated inward and outward, respectively.

and solution to KL loss at each diffusion step t based on the latent mixture model.

III. CONTRASTIVE MIXTURE DIFFUSION

This study deals with the insufficiency in previous diffusion models [3], [9] and presents a new Gaussian-Dirac mixture diffusion process where a continuous-discrete denoising network f_θ is estimated to implement a mixture diffusion as shown in Fig. 2. A contrastive learning [30] is merged to carry out a *contrastive mixture diffusion* model, which reveals the easy-first strategy for text generation, as detailed below.

A. Easy-First Text Generation

Traditionally, text generation was operated through an autoregressive (AR) language model where each token in a sentence was predicted one by one in a left-to-right manner. This AR LM is basically difficult to perform long-range planning for conditional sequence generation [31]. Importantly, the modern language models are operated in a non-AR way where the generative diffusion model was developed [2]. Fig. 3(a) illustrates a mask language model, which refines and denoises a string from mask tokens $\mathbf{m}$ or $[\mathbf{M}]$ towards its corresponding clean string in reverse process, which follows the easy-first generation strategy. Each diffusion step refines the generated string in embedding space by a continuous-discrete denoising process driven by (12). The easy or common tokens “The” and “on” are generated in earlier step relative to the specific tokens “sat”

and “mat” generated in later step. There are two stages in the generative model. The first stage is to build a mixture model to accommodate mask tokens in a continuous diffusion model while the second stage is to introduce the contrastive mixture diffusion to fulfill easy-first generation in the training process. This study builds a contrastive Gaussian-Dirac mixture model for continuous diffusion, leveraged by the ordinary differential equation solver [7] for efficient computation. The trained model likely follows the easy-first generation in inference phase, which is beneficial for system performance.

B. Contrastive Diffusion Model

This work in Fig. 2 further introduces contrastive learning with mask embedding $\mathbf{m}$ as the anchor to enhance the easy-first behavior in diffusion-based generation. In particular, an embedding encoder g_ϕ is learned to build a contrastive embedding space, which is seamlessly merged with the estimation of diffusion denoiser f_θ in the proposed Gaussian-Dirac mixture diffusion. In addition to the diffusion loss $\mathcal{L}_{diff}$ in (9) consisting of the reconstruction loss $\mathcal{L}_{rc}$ and the alignment loss $\mathcal{L}_{kl} = \sum_{t=2}^T \mathcal{L}_{t-1}$, an additional triplet loss for contrastive embedding is incorporated in training the proposed embedder-denoiser diffusion network. All samples are located in the embedding space for diffusion, which is constructed by the embedding layer g_ϕ by minimizing [32]

$$\mathcal{L}_{tri} = \sum_{(\mathbf{p}, \mathbf{n}) \in P \times N} \max(\|\mathbf{m} - \mathbf{p}\|^2 - \|\mathbf{m} - \mathbf{n}\|^2 + \xi, 0) \quad (13)$$

where mask token $\mathbf{m}$ is the anchor, ξ is the margin in metric learning, and $P \times N = \{\mathbf{p}^{(i)}, \mathbf{n}^{(i)}\}_{i=1}^M$ denotes a set of M pairs of positive $\mathbf{p}^{(i)}$ and negative $\mathbf{n}^{(i)}$ samples for contrastive learning [33]. The learned embedding space is spanned as a kind of concentric circles, centered around the mask embedding $\mathbf{m}$, with the number of word occurrences decreasing from the center to the outward region. This triplet loss is always positive and is minimized to carry out rank ordering via the distances of different sample embeddings $\mathbf{z}_t^n$ relative to mask embedding $\mathbf{m}$. As a result, the positive sample is not closer than the negative sample by at least a margin ξ [34]. In that case, the triplet loss minimization will give the positive sample $\mathbf{p}^{(i)}$ (like the words “A” and “The”) an update signal to pull it towards the anchor sample $\mathbf{m}$, and simultaneously the negative sample $\mathbf{n}^{(i)}$ (like the words “loss” and “optimize” in Fig. 2) will receive an update signal to push it far away from the anchor $\mathbf{m}$. Minimizing $\mathcal{L}_{tri}$ ensures that the generation process starts from the mask token and moves outward, passing through the easier token region with higher word frequency before reaching the harder token region with lower word frequency, thus fulfilling the easy-first property for text generation. The latent diffusion model is learned by using the sample embeddings $\mathbf{z}_t^n$ based on an encoder g_ϕ , which is estimated according to $\mathcal{L}_{tri}$. Fig. 3(b) illustrates how the set of paired positive and negative word embeddings is selected according to a random split s . Basically, the set of all vocabulary words is first ranked in a decreasing order based on the number of word occurrences. The ordered vocabulary list is split randomly into two groups at each training iteration. The group of words

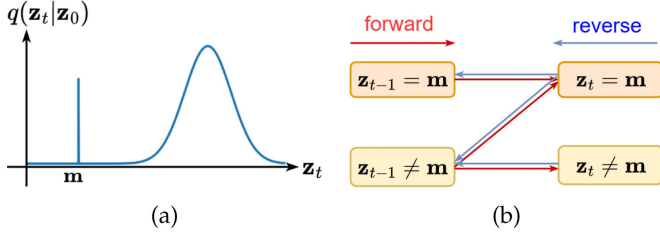


Fig. 4. (a) Gaussian-Dirac mixture distribution. (b) Absorbing mask token $\mathbf{m}$ in forward and reverse processes.

with higher frequency is considered as positive samples $\mathbf{p}^{(i)}$, and that with lower frequency is seen as negative samples $\mathbf{n}^{(i)}$. The triplet loss is accumulated for all possible positive-negative pairs $(\mathbf{p}^{(i)}, \mathbf{n}^{(i)}) \sim P \times N$.

C. Gaussian-Dirac Mixture Diffusion

This study formulates a new framework for sequence diffusion where the *probabilistic continuous-discrete noise* based on (12) and (11) is truthfully expressed to yield the forward transition distribution. As depicted in Fig. 4(a), a mixture distribution [35], [36] is formed to bridge between original sample $\mathbf{z}_0^n$ and noisy sample $\mathbf{z}_t^n$ at each step t via

$$\begin{aligned} \mathbf{z}_t^n &\sim q(\mathbf{z}_t^n | \mathbf{z}_0^n) \\ &= b_t \delta(\mathbf{z}_t^n - \mathbf{m}) + (1 - b_t) \mathcal{N}(\mathbf{z}_t^n; \sqrt{\bar{\alpha}_t} \mathbf{z}_0^n, (1 - \bar{\alpha}_t) \mathbf{I}) \end{aligned} \quad (14)$$

which is seen as a mixture of Dirac delta function and Gaussian diffusion distribution. In (14), $0 \leq b_t \leq 1$ is a mixture weight, and $\mathbf{m}$ is a learnable soft mask controlled by a parameter γ_t for a random mask indicator ρ_t^n in (12). In diffusion process, each token $\mathbf{z}_t^n$ at position n of a string $\mathbf{z}_t$ is drawn with a *mixture absorbing noise*. Fig. 4(b) illustrates the mixture diffusion transition in forward and reverse processes in presence of absorbing mask token $\mathbf{m}$. Forward process $\mathbf{z}_{t-1} \rightarrow \mathbf{z}_t$ will regress towards the absorbing state $\mathbf{z}_t = \mathbf{m}$ while reverse process $\mathbf{z}_t \rightarrow \mathbf{z}_{t-1}$ will denoise towards deviating from $\mathbf{m}$ via $\mathbf{z}_{t-1} \neq \mathbf{m}$.

In particular, this paper presents a *Gaussian-Dirac mixture diffusion* model based on the mixture transition distribution $q(\mathbf{z}_t^n | \mathbf{z}_0^n)$ in (14), which necessitates two calculations for

- forward posterior $q(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n, \mathbf{z}_0^n)$, and
- KL divergence in alignment loss $\mathcal{L}_{t-1}$ between forward and reverse processes at each diffusion step t ,

to ensure that the new process is accurately represented and the theoretical framework remains consistent.

For the first calculation, we refer to (5) and calculate the forward posterior via $q(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n, \mathbf{z}_0^n) \propto q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n, \mathbf{z}_0^n) q(\mathbf{z}_{t-1}^n | \mathbf{z}_0^n)$. Considering the Markov property $q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n, \mathbf{z}_0^n) = q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n)$ and the absorbing property for soft token $\mathbf{m}$, the one-step forward transition under a continuous-discrete noise process is driven by

$$q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n) = \begin{cases} \delta(\mathbf{z}_t^n - \mathbf{m}), & \text{if } \mathbf{z}_{t-1}^n = \mathbf{m} \\ w_1 \mathcal{N}(\mathbf{z}_t^n; \mathbf{m}, \sigma^2 \mathbf{I}) + w_0 \\ \quad \times \mathcal{N}(\mathbf{z}_t^n; \sqrt{\bar{\alpha}_t} \mathbf{z}_{t-1}^n, (1 - \bar{\alpha}_t) \mathbf{I}), & \text{if } \mathbf{z}_{t-1}^n \neq \mathbf{m} \end{cases} \quad (15)$$

where w_1 and w_0 denote the weights for discrete and continuous noises, respectively. This transition process is constructed by checking whether the previous embedding $\mathbf{z}_{t-1}^n$ is associated with a mask token $\mathbf{m}$ or not. Notably, Gaussian-Dirac mixture distribution is expressed by either the absorbing noise $\mathbf{z}_t^n = \mathbf{m}$ or a mixture of Gaussians where a Gaussian $\mathcal{N}(\mathbf{z}_t^n; \mathbf{m}, \sigma^2 \mathbf{I})$ with an arbitrary small $\sigma^2 \rightarrow 0$ is used to approximate the Dirac delta function $\delta(\mathbf{z}_t^n - \mathbf{m})$.

There are two cases in (15), which are incorporated in finding the forward posterior distribution. First, the case of $\mathbf{z}_{t-1}^n = \mathbf{m}$ is considered to obtain

$$\begin{aligned} q(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n, \mathbf{z}_0^n) &\propto q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n = \mathbf{m}, \mathbf{z}_0^n) q(\mathbf{z}_{t-1}^n = \mathbf{m} | \mathbf{z}_0^n) \\ &= b_{t-1} \delta(\mathbf{z}_t^n - \mathbf{m}). \end{aligned} \quad (16)$$

Considering the alternative case $\mathbf{z}_{t-1}^n \neq \mathbf{m}$, the corresponding forward posterior is derived by finding $q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n \neq \mathbf{m}, \mathbf{z}_0^n)$ and $q(\mathbf{z}_{t-1}^n \neq \mathbf{m} | \mathbf{z}_0^n)$, which are obtained by applying Markov property in

$$\begin{aligned} q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n, \mathbf{z}_0^n) &= q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n) \\ &= w_1 \mathcal{N}(\mathbf{z}_t^n; \mathbf{m}, \sigma^2 \mathbf{I}) \\ &\quad + w_0 \mathcal{N}(\mathbf{z}_t^n; \sqrt{\bar{\alpha}_t} \mathbf{z}_{t-1}^n, (1 - \bar{\alpha}_t) \mathbf{I}) \end{aligned} \quad (17)$$

and by referring (14) to yield

$$\begin{aligned} q(\mathbf{z}_{t-1}^n | \mathbf{z}_0^n) &= b_{t-1} \mathcal{N}(\mathbf{z}_{t-1}^n; \mathbf{m}, \sigma^2 \mathbf{I}) \\ &\quad + (1 - b_{t-1}) \mathcal{N}(\mathbf{z}_{t-1}^n; \sqrt{\bar{\alpha}_{t-1}} \mathbf{z}_0^n, (1 - \bar{\alpha}_{t-1}) \mathbf{I}). \end{aligned} \quad (18)$$

Therefore, the forward posterior is calculated by multiplying individual two terms in right-hand-side (RHS) of (17) and (18) to find the following four terms

$$\begin{aligned} q(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n, \mathbf{z}_0^n) &\propto w_1 b_{t-1} \mathcal{N}(\mathbf{z}_{t-1}^n; \mathbf{m}, \sigma^2 \mathbf{I}) \mathcal{N}(\mathbf{z}_t^n; \mathbf{m}, \sigma^2 \mathbf{I}) \\ &\quad + w_0 b_{t-1} \mathcal{N}(\mathbf{z}_{t-1}^n; \mathbf{m}, \sigma^2 \mathbf{I}) \mathcal{N}(\mathbf{z}_t^n; \sqrt{\bar{\alpha}_t} \mathbf{z}_{t-1}^n, (1 - \bar{\alpha}_t) \mathbf{I}) \\ &\quad + w_1 (1 - b_{t-1}) \left(\mathcal{N}(\mathbf{z}_{t-1}^n; \sqrt{\bar{\alpha}_{t-1}} \mathbf{z}_0^n, (1 - \bar{\alpha}_{t-1}) \mathbf{I}) \right. \\ &\quad \times \mathcal{N}(\mathbf{z}_t^n; \mathbf{m}, \sigma^2 \mathbf{I}) \\ &\quad \left. + w_0 (1 - b_{t-1}) \left(\mathcal{N}(\mathbf{z}_{t-1}^n; \sqrt{\bar{\alpha}_{t-1}} \mathbf{z}_0^n, (1 - \bar{\alpha}_{t-1}) \mathbf{I}) \right. \right. \\ &\quad \times \mathcal{N}(\mathbf{z}_t^n; \sqrt{\bar{\alpha}_t} \mathbf{z}_{t-1}^n, (1 - \bar{\alpha}_t) \mathbf{I}) \left. \left. \right) \right). \end{aligned} \quad (19)$$

In (19), we may arrange a posterior distribution of $\mathbf{z}_{t-1}^n$ given by condition $\mathbf{z}_t^n$ and observation $\mathbf{z}_0^n$. So, the terms that do not depend on $\mathbf{z}_{t-1}^n$ are treated as the constants. To do so, a Gaussian distribution of $\mathbf{z} \in \mathbb{R}^D$ is rewritten by

$$\mathcal{N}(\mathbf{z}; \lambda \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\lambda^D} \mathcal{N}\left(\boldsymbol{\mu}; \frac{\mathbf{z}}{\lambda}, \frac{\boldsymbol{\Sigma}}{\lambda^2}\right). \quad (20)$$

Therefore, $\mathcal{N}(\mathbf{z}_t^n; \sqrt{\bar{\alpha}_t} \mathbf{z}_{t-1}^n, (1 - \bar{\alpha}_t) \mathbf{I})$ in the fourth term of (19) can be rearranged as

$$\frac{1}{(\sqrt{\bar{\alpha}_t})^D} \mathcal{N}\left(\mathbf{z}_{t-1}^n; \frac{\mathbf{z}_t^n}{\sqrt{\bar{\alpha}_t}}, (\bar{\alpha}_t^{-1} - 1) \mathbf{I}\right). \quad (21)$$

Then, using the formula of rearranging the product of two Gaussian distributions of $\mathbf{z}$ in a form of

$$\mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_a, \sigma_a^2 \mathbf{I}) \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_b, \sigma_b^2 \mathbf{I}) = \mathcal{N}(\boldsymbol{\mu}_a; \boldsymbol{\mu}_b, (\sigma_a^2 + \sigma_b^2) \mathbf{I}) \times \mathcal{N}\left(\mathbf{z}; \frac{\frac{1}{\sigma_a^2} \boldsymbol{\mu}_a + \frac{1}{\sigma_b^2} \boldsymbol{\mu}_b}{\frac{1}{\sigma_a^2} + \frac{1}{\sigma_b^2}}, \frac{1}{\frac{1}{\sigma_a^2} + \frac{1}{\sigma_b^2}} \mathbf{I}\right) \quad (22)$$

the resulting equation becomes only one Gaussian for $\mathbf{z}$ since the other Gaussian is independent of $\mathbf{z}$. Applying (22), the products of Gaussians in (19) become

$$\begin{aligned} & \mathcal{N}(\mathbf{z}_{t-1}^n; \mathbf{m}, \sigma^2 \mathbf{I}) \mathcal{N}(\mathbf{z}_t^n; \sqrt{\alpha_t} \mathbf{z}_{t-1}^n, (1 - \alpha_t) \mathbf{I}) \\ &= \frac{1}{(\sqrt{\alpha_t})^D} \mathcal{N}\left(\frac{\mathbf{z}_t^n}{\sqrt{\alpha_t}}; \mathbf{m}, \left(\sigma^2 + \frac{1 - \alpha_t}{\alpha_t}\right) \mathbf{I}\right) \mathcal{N}(\mathbf{z}_{t-1}^n; \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2) \end{aligned} \quad (23)$$

$$\begin{aligned} & \mathcal{N}(\mathbf{z}_{t-1}^n; \sqrt{\bar{\alpha}_{t-1}} \mathbf{z}_0^n, (1 - \bar{\alpha}_{t-1}) \mathbf{I}) \mathcal{N}(\mathbf{z}_t^n; \sqrt{\alpha_t} \mathbf{z}_{t-1}^n, (1 - \alpha_t) \mathbf{I}) \\ &= \frac{1}{(\sqrt{\alpha_t})^D} \mathcal{N}\left(\frac{\mathbf{z}_t^n}{\sqrt{\alpha_t}}; \frac{\sqrt{\bar{\alpha}_t} \mathbf{z}_0^n}{\sqrt{\alpha_t}}, \frac{1 - \bar{\alpha}_t}{\alpha_t} \mathbf{I}\right) \mathcal{N}(\mathbf{z}_{t-1}^n; \boldsymbol{\mu}_4, \boldsymbol{\Sigma}_4). \end{aligned} \quad (24)$$

Finally, the forward posterior is derived as a mixture of four Gaussians as

$$q(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n, \mathbf{z}_0^n) \propto \sum_{k=1}^4 c_k \mathcal{N}(\mathbf{z}_{t-1}^n; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad (25)$$

where the mixture weights are yielded by

$$\begin{aligned} c_1 &= w_1 b_{t-1} \mathcal{N}(\mathbf{z}_t^n; \mathbf{m}, \sigma^2 \mathbf{I}) \\ c_2 &= w_0 b_{t-1} \frac{1}{(\sqrt{\alpha_t})^D} \mathcal{N}\left(\frac{\mathbf{z}_t^n}{\sqrt{\alpha_t}}; \mathbf{m}, \left(\sigma^2 + \frac{1 - \alpha_t}{\alpha_t}\right) \mathbf{I}\right) \\ c_3 &= w_1 (1 - b_{t-1}) \mathcal{N}(\mathbf{z}_t^n; \mathbf{m}, \sigma^2 \mathbf{I}) \\ c_4 &= w_0 (1 - b_{t-1}) \frac{1}{(\sqrt{\alpha_t})^D} \mathcal{N}\left(\frac{\mathbf{z}_t^n}{\sqrt{\alpha_t}}; \frac{\sqrt{\bar{\alpha}_t} \mathbf{z}_0^n}{\sqrt{\alpha_t}}, \frac{1 - \bar{\alpha}_t}{\alpha_t} \mathbf{I}\right) \end{aligned} \quad (26)$$

and the mixture means and variances are given by

$$\begin{aligned} \boldsymbol{\mu}_1 &= \mathbf{m}, & \boldsymbol{\Sigma}_1 &= \sigma^2 \mathbf{I} \\ \boldsymbol{\mu}_2 &= \frac{(1 - \alpha_t) \mathbf{m} + \sigma^2 \sqrt{\alpha_t} \mathbf{z}_t^n}{1 - \alpha_t + \sigma^2 \alpha_t}, & \boldsymbol{\Sigma}_2 &= \frac{1}{(1 - \alpha_t)^{-1} \alpha_t + \sigma^2} \mathbf{I} \\ \boldsymbol{\mu}_3 &= \sqrt{\bar{\alpha}_{t-1}} \mathbf{z}_0^n, & \boldsymbol{\Sigma}_3 &= (1 - \bar{\alpha}_{t-1}) \mathbf{I} \\ \boldsymbol{\mu}_4 &= \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1}) \mathbf{z}_t^n + \sqrt{\bar{\alpha}_{t-1}} (1 - \alpha_t) \mathbf{z}_0^n}{1 - \bar{\alpha}_t}, & \boldsymbol{\Sigma}_4 &= (1 - \bar{\alpha}_t)^{-1} \mathbf{I} \end{aligned} \quad (27)$$

where $\{\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2\}$ and $\{\boldsymbol{\mu}_4, \boldsymbol{\Sigma}_4\}$ in (23) and (24) are derived by referring the second Gaussian of the formula in RHS of (22), respectively.

Interestingly, the Gaussian-Dirac mixture distribution in (14) for forward diffusion can be reformulated to connect the relation between the mixture weights $\{b_t, 1 - b_t\}$ in (14) and $\{w_0, w_1\}$ and in (15), respectively. To do so, firstly, we consider both cases $\{\mathbf{z}_{t-1}^n = \mathbf{m}\}$ and $\{\mathbf{z}_{t-1}^n \neq \mathbf{m}\}$ in finding the predictive distribution in the first and second terms in RHS of the following

equation

$$\begin{aligned} q(\mathbf{z}_t^n | \mathbf{z}_0^n) &= \int_{\mathbb{R}^D} q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n, \mathbf{z}_0^n) q(\mathbf{z}_{t-1}^n | \mathbf{z}_0^n) d\mathbf{z}_{t-1}^n \\ &= b_{t-1} q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n = \mathbf{m}, \mathbf{z}_0^n) + (1 - b_{t-1}) \\ &\quad w_1 \delta(\mathbf{z}_t^n - \mathbf{m}) \\ &\quad \times \left[\int_{\{\mathbf{z}_{t-1}^n \neq \mathbf{m}\}} \times \mathcal{N}(\mathbf{z}_{t-1}^n; \sqrt{\bar{\alpha}_{t-1}} \mathbf{z}_0^n, (1 - \bar{\alpha}_{t-1}) \mathbf{I}) d\mathbf{z}_{t-1}^n \right. \\ &\quad \left. w_0 \mathcal{N}(\mathbf{z}_t^n; \sqrt{\alpha_t} \mathbf{z}_{t-1}^n, (1 - \alpha_t) \mathbf{I}) \right. \\ &\quad \left. + \int_{\{\mathbf{z}_{t-1}^n \neq \mathbf{m}\}} \times \mathcal{N}(\mathbf{z}_{t-1}^n; \sqrt{\bar{\alpha}_{t-1}} \mathbf{z}_0^n, (1 - \bar{\alpha}_{t-1}) \mathbf{I}) d\mathbf{z}_{t-1}^n \right] \end{aligned} \quad (28)$$

respectively, where $q(\mathbf{z}_t^n | \mathbf{z}_{t-1}^n, \mathbf{z}_0^n)$ in (17) and $q(\mathbf{z}_{t-1}^n | \mathbf{z}_0^n)$ in (18) are merged, and the first term is obtained due to a delta function in the integral. In RHS of (28), the first term is obtained by applying the absorbing property at $\mathbf{z}_t = \mathbf{m}$, and inside the brackets of the second term, the first term is derived due to an integral over a Gaussian of $\mathbf{z}_{t-1}^n$ being one and the second term is yielded by applying (22) for integrating over the multiplication of two Gaussians of $\mathbf{z}_{t-1}^n$. The resulting distribution after this arrangement is found as

$$\begin{aligned} q(\mathbf{z}_t^n | \mathbf{z}_0^n) &= b_{t-1} \delta(\mathbf{z}_t^n - \mathbf{m}) + (1 - b_{t-1}) w_1 \delta(\mathbf{z}_t^n - \mathbf{m}) \\ &\quad + (1 - b_{t-1}) w_0 \mathcal{N}(\mathbf{z}_t^n; \sqrt{\bar{\alpha}_t} \mathbf{z}_0^n, (1 - \bar{\alpha}_t) \mathbf{I}) \\ &= (b_{t-1} + (1 - b_{t-1}) w_1) \delta(\mathbf{z}_t^n - \mathbf{m}) \\ &\quad + (1 - b_{t-1}) w_0 \mathcal{N}(\mathbf{z}_t^n; \sqrt{\bar{\alpha}_t} \mathbf{z}_0^n, (1 - \bar{\alpha}_t) \mathbf{I}). \end{aligned} \quad (29)$$

Therefore, w_0 and w_1 can be solved by equating the coefficients in the Gaussian-Dirac distributions in (29) and (14), which yield the following system of equations

$$\begin{cases} w_1 + w_0 = 1 \\ (b_{t-1} + (1 - b_{t-1}) w_1) + (1 - b_{t-1}) w_0 = 1 \\ b_{t-1} + (1 - b_{t-1}) w_1 = b_t \\ (1 - b_{t-1}) w_0 = 1 - b_t \end{cases} \quad (30)$$

where the mixture weights of a mixture density summed up to 1 is applied. Then, a unique solution is derived as

$$w_0 = \frac{1 - b_t}{1 - b_{t-1}}, \quad w_1 = \frac{b_t - b_{t-1}}{1 - b_{t-1}}. \quad (31)$$

The final step is to utilize the property $\sigma^2 \rightarrow 0$ for the case $\mathbf{z}_{t-1}^n = \mathbf{m}$ in (25), (26) and (27). It is found that the first two Gaussians from a mixture of four Gaussians are vanished to a Dirac delta function

$$\{\mathcal{N}(\mathbf{z}_{t-1}^n; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1), \mathcal{N}(\mathbf{z}_{t-1}^n; \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)\} \rightarrow \delta(\mathbf{z}_{t-1}^n - \mathbf{m}). \quad (32)$$

The Gaussians of $\mathbf{z}_t^n$ in the terms c_1 and c_2 become one. The case of $\mathbf{z}_{t-1}^n \neq \mathbf{m}$ results in the mixture of the third and fourth Gaussians. Equation (25) is then rewritten by

$$q(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n, \mathbf{z}_0^n) \propto \begin{cases} b_{t-1} \delta(\mathbf{z}_{t-1}^n - \mathbf{m}), & \text{if } \mathbf{z}_t^n = \mathbf{m} \\ \sum_{k=3}^4 c_k \mathcal{N}(\mathbf{z}_{t-1}^n; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), & \text{if } \mathbf{z}_t^n \neq \mathbf{m}. \end{cases} \quad (33)$$

Looking carefully at (33), (26) and (27), it is found that the third Gaussian mixture becomes negligible since this term holds value only when $\mathbf{z}_t^n = \mathbf{m}$ due to the Dirac delta function $\delta(\mathbf{z}_t^n - \mathbf{m})$, but this is controversial with the condition $\mathbf{z}_t^n \neq \mathbf{m}$ in (33). As a result, only the fourth Gaussian mixture exists in (33), which produces a mixture distribution in an approximate form consisting of two Gaussians to represent a kind of continuous-discrete noising (or denoising) process as

$$q(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n, \mathbf{z}_0^n) \propto b_{t-1} \mathcal{N}(\mathbf{z}_{t-1}^n; \mathbf{m}, \sigma^2 \mathbf{I}) + c_4 \mathcal{N}(\mathbf{z}_{t-1}^n; \boldsymbol{\mu}_4, \boldsymbol{\Sigma}_4). \quad (34)$$

In (34), the first Gaussian is to indicate a discrete noise at a learnable mask token $\mathbf{m}$ weighted by b_{t-1} with $\sigma^2 \rightarrow 0$ while the second Gaussian is to reveal a continuous Gaussian noise located at $\boldsymbol{\mu}_4$ with scaling parameters $\boldsymbol{\Sigma}_4$ and c_4 . Importantly, this new sequence diffusion is driven by a denoising Gaussian-Dirac distribution, which is theoretically derived by combining the mask token $\mathbf{m}$ with an interpolation of previous noisy sample $\mathbf{z}_t^n$ and estimated clean sample $\hat{\mathbf{z}}_0^n$, as shown in $\boldsymbol{\mu}_4$ given in (27).

For the second calculation, we need to derive the KL divergence or alignment loss $\mathcal{L}_{t-1}$ between the transition distributions $q(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n, \mathbf{z}_0^n)$ in forward process and $p_\theta(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n)$ in reverse process at each diffusion step t . Consequently, this study refers to (34) and introduces a reverse transition distribution with a denoising parameter θ in a form of a mixture of two Gaussians with weights $\{w_d^p, w_g^p\}$ as

$$p_\theta(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n) \propto w_d^p \mathcal{N}(\mathbf{z}_{t-1}^n; \boldsymbol{\mu}_\theta(\mathbf{m}, t), \sigma^2 \mathbf{I}) + w_g^p \mathcal{N}(\mathbf{z}_{t-1}^n; \boldsymbol{\mu}_\theta(\mathbf{z}_t^n, t), (1 - \bar{\alpha}_t)^{-1} \mathbf{I}). \quad (35)$$

The denoising network f_θ is learned to simulate the Gaussian means $\boldsymbol{\mu}_\theta(\cdot, t)$ for discrete noise and continuous noise by feeding in mask token $\mathbf{m}$ and current noisy token $\mathbf{z}_t^n$, respectively. The Gaussian variances in two processes are unchanged. Correspondingly, this work treats the output of a denoiser $\hat{\mathbf{z}}_0(\cdot, t)$ as the estimated clean sample $\hat{\mathbf{z}}_0$ and then substitutes it into $\boldsymbol{\mu}_4$ in (27) or $\tilde{\boldsymbol{\mu}}_t(\cdot, \mathbf{z}_0)$ in (6) to find $\boldsymbol{\mu}_\theta(\cdot, t) \leftarrow \tilde{\boldsymbol{\mu}}_t(\cdot, \hat{\mathbf{z}}_0)$ for Gaussian means in (35). Therefore, $q(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n, \mathbf{z}_0^n)$ and $p_\theta(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n)$ are both seen as a linear combination of two Gaussian distributions.

D. Implementation and Algorithm

In the implementation, the KL divergence between two linear combinations of Gaussians in $\mathcal{L}_{\text{kl}}$ still has no analytical form. This study leverages a compromise solution to approximate KL divergence between any p and q , caused by the mixture of Dirac and Gaussian distributions in a form of

$$q = w_d^q q_d + w_g^q q_g, \quad p = w_d^p p_d + w_g^p p_g. \quad (36)$$

These mixtures of Gaussians are driven by the mixture weights w_d^q, w_g^q, w_d^p and w_g^p for the Gaussians q_d, q_g, p_d and p_g , respectively, which could be correspondingly obtained by (34) and (35). Then, KL divergence can be approximated by an average of upper and lower bounds [37], [38]

$$\mathcal{D}_{\text{kl}}(q \| p_\theta) \approx \frac{1}{2} (\mathcal{D}_{\text{prod}}(q \| p_\theta) + \mathcal{D}_{\text{var}}(q \| p_\theta)) \quad (37)$$

based on the product of Gaussian approximation and the variational approximation, given by

$$\mathcal{D}_{\text{prod}}(q \| p_\theta) = \sum_i w_i^q \log \frac{\sum_j w_j^q \zeta_{ij}}{\sum_k w_k^p \tau_{ik}} \quad (38)$$

$$\mathcal{D}_{\text{var}}(q \| p_\theta) = \sum_i w_i^q \log \frac{\sum_j w_j^q \exp(-\mathcal{D}_{\text{kl}}(q_i \| q_j))}{\sum_k w_k^p \exp(-\mathcal{D}_{\text{kl}}(q_i \| p_{\theta,k}))} \quad (39)$$

respectively, $i, j, k \in \{d, g\}$ includes Dirac and Gaussian, and

$$\zeta_{ij} \triangleq \int q_i(\mathbf{z}) q_j(\mathbf{z}) d\mathbf{z}, \quad \tau_{ij} \triangleq \int q_i(\mathbf{z}) p_{\theta,j}(\mathbf{z}) d\mathbf{z}. \quad (40)$$

KL divergence between two Gaussian mixtures in (37) is derived with the divergence and the integral over individual Gaussians in $\mathcal{D}_{\text{kl}}(q_i \| q_j)$, $\mathcal{D}_{\text{kl}}(q_i \| p_{\theta,k})$, ζ_{ij} and τ_{ij} .

Typically, the function $\mathcal{D}_{\text{prod}}$ tends to overestimate KL divergence $\mathcal{D}_{\text{kl}}$ while $\mathcal{D}_{\text{var}}$ tends to underestimate. The difference of them represents twice of the maximum of the error between real $\mathcal{D}_{\text{kl}}$ and its approximation to $\mathcal{L}_{\text{kl}}$ [37]. For an empirical consideration, a threshold η is selected to disregard those imprecise KL approximations from different samples $\mathbf{z}_t$ at different diffusion steps t . The KL loss for the proposed contrastive mixture diffusion (also denoted by CMD) is implemented via an indicator function $\mathbb{1}(\cdot)$ by

$$\mathcal{L}_{\text{kl}} = \sum_{t=2}^T \mathbb{E}_{q(\mathbf{z}_t | \mathbf{z}_0)} [\mathbb{1}(|\mathcal{D}_{\text{prod}}(q_t \| p_{\theta,t}) - \mathcal{D}_{\text{var}}(q_t \| p_{\theta,t})| < \eta) \mathcal{D}_{\text{kl}}(q_t \| p_{\theta,t})] \quad (41)$$

where only the approximations in (38) and (39) with significant closeness are included. In addition to KL loss $\mathcal{L}_{\text{kl}}$, the diffusion objective $\mathcal{L}_{\text{diff}}$ aims to learn the denoiser parameter θ also involves the reconstruction loss

$$\mathcal{L}_{\text{rc}} \triangleq -\mathbb{E}_{q(\mathbf{z}_1 | \mathbf{z}_0)} [\log p_\theta(\mathbf{z}_0 | \mathbf{z}_1)] \propto \mathbb{E}_{q(\mathbf{z}_1 | \mathbf{z}_0)} [\|\mathbf{z}_0 - \hat{\mathbf{z}}_0(\mathbf{z}_1)\|^2] \quad (42)$$

which is obtained as a mean squared error due to a standard Gaussian for $\mathbf{z}_0 - \hat{\mathbf{z}}_0(\mathbf{z}_1)$ in a likelihood function $p_\theta(\mathbf{z}_0 | \mathbf{z}_1)$. Notably, the denoised output $\hat{\mathbf{z}}_0$, given by $\mathbf{z}_1$, is gradually diffused from $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ at different steps t . The overall learning objectives for *contrastive embedder* g_ϕ and *diffusion denoiser* f_θ in the CMD model are finally formed by

$$\mathcal{L}(\phi, \theta, \mathbf{m}) = \lambda_c \mathcal{L}_{\text{tri}}(\phi) + \lambda_a \mathcal{L}_{\text{kl}}(\phi, \theta, \mathbf{m}) + \lambda_r \mathcal{L}_{\text{rc}}(\phi, \theta, \mathbf{m}) \quad (43)$$

where λ_c , λ_a and λ_r denote the regularization parameters for contrastive learning, transition alignment and reconstruction matching, respectively.

Algorithm 1 illustrates the proposed CMD training procedure for embedder ϕ , soft mask $\mathbf{m}$ and denoiser θ . First of all, the condition and target texts from dataset $\mathcal{D} = \{\mathbf{x}_0, \mathbf{y}_0\}$ are concatenated and then passed the concatenation through the embedding or encoding layer g_ϕ , resulting in the embedding $\mathbf{z}_0$. The embedding layer is seen as a learnable matrix ϕ , which maps between a token identity and its embedding vector. This matrix is ordered according to the number of word occurrences from high frequency to low frequency. Diffusion model is implemented

Algorithm 1: CMD Training Procedure.

Require : dataset $\mathcal{D}$, sorted embed layer ϕ , diffusion denoiser θ , soft mask $\mathbf{m}$, triplet margin ξ , mask schedule $\{\gamma_t\}$, error tolerance η

repeat

 sample condition and target texts $(\mathbf{x}_0, \mathbf{y}_0) \sim \mathcal{D}$

 map tokens to embeddings $\mathbf{z}_0 = g_\phi(\mathbf{x}_0 \oplus \mathbf{y}_0)$

 sample a diffusion step $t \sim \mathcal{U}[1, T]$

for each token n **do**

 sample $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

 sample $\rho_t^n \sim \text{Bernoulli}(\gamma_t)$

$\mathbf{z}_t^n = \begin{cases} \mathbf{m}, & \text{if } \rho_t^n = 1 \\ \sqrt{\bar{\alpha}_t} \mathbf{z}_0^n + \sqrt{1 - \bar{\alpha}_t} \epsilon, & \text{if } \rho_t^n = 0 \end{cases}$

end

 sample a positive-negative split $s \sim \mathcal{U}[1, |\phi| - 2]$

 define positive samples $P \leftarrow \phi[0 : s]$

 define negative samples $N \leftarrow \phi[s + 1 : |\phi| - 1]$

 initialize via $\mathcal{L}_{\text{tri}} \leftarrow 0$

 apply stopgrad($\mathbf{m}$)

for each pair $(\mathbf{p}, \mathbf{n}) \sim P \times N$ **do**

$\mathcal{L}_{\text{tri}} \leftarrow \mathcal{L}_{\text{tri}} + \max(\|\mathbf{m} - \mathbf{p}\|^2 - \|\mathbf{m} - \mathbf{n}\|^2 + \xi, 0)$

end

 predict a denoiser output $\hat{\mathbf{z}}_0 = f_\theta(\mathbf{z}_t, t)$

 accumulate $\mathcal{L}_{\text{rc}}$ using $\|\mathbf{z}_0 - \hat{\mathbf{z}}_0\|^2$ in (42)

 calculate mixture weights in (18)(31)(34)(35)

 calculate $\mathcal{D}_{\text{prod}}$ in (38) and $\mathcal{D}_{\text{var}}$ in (39)

 approximate $\mathcal{D}_{\text{kl}}$ by (37)

 calculate $\mathcal{L}_{\text{kl}}$ by (41) with η

 update ϕ by descent on $\nabla_\phi(\mathcal{L}_{\text{tri}} + \mathcal{L}_{\text{kl}} + \mathcal{L}_{\text{rc}})$

 update θ by descent on $\nabla_\theta(\mathcal{L}_{\text{kl}} + \mathcal{L}_{\text{rc}})$

 update $\mathbf{m}$ by descent on $\nabla_{\mathbf{m}}(\mathcal{L}_{\text{kl}} + \mathcal{L}_{\text{rc}})$

until converged;

in the latent space by using clean embeddings $\mathbf{z}_0$. Then, a diffusion step is sampled from a uniform distribution $\mathcal{U}$ between 1 and T . The target portion of $\mathbf{z}_0$ undergoes a noising process that combines continuous Gaussian noise signal and discrete Dirac mask token. A noise ϵ is drawn from a standard Gaussian and added to each token embedding $\mathbf{z}_t^n$ in the sentence $\mathbf{z}_0$. ρ_t^n indicates whether the embedding $\mathbf{z}_t^n$ is masked by $\mathbf{m}$. If $\mathbf{z}_t^n$ is not masked, it undergoes a regular Gaussian noise addition. In calculation of triplet loss $\mathcal{L}_{\text{tri}}$, a random split point s is first drawn from $\mathcal{U}[1, |\phi| - 2]$ to divide the embeddings ϕ with size $|\phi|$ into the positive samples in P and negative samples in N . The initial condition is set by $\mathcal{L}_{\text{tri}} \leftarrow 0$. Since we do not want the triplet loss to update mask embedding $\mathbf{m}$, the trick to stop gradient stopgrad($\mathbf{m}$) is used to ensure that $\mathbf{m}$ is excluded from gradient calculation. The Cartesian product $P \times N$ is then formed to consider all possible combinations of $\mathbf{p} \in P$ and $\mathbf{n} \in N$. The triplet loss $\mathcal{L}_{\text{tri}}$ is calculated and accumulated from individual combinations, and is specially designed to regulate the embedding layer g_ϕ . Although this loss involves soft mask $\mathbf{m}$, the gradient calculation was empirically treated by neglecting its dependency with $\mathbf{m}$. Next, the noisy sample $\mathbf{z}_t$ is fed into the denoiser f_θ to predict $\hat{\mathbf{z}}_0$. The objective $\mathcal{L}_{\text{rc}}$ is accumulated from the squared error between $\hat{\mathbf{z}}_0$ and its original $\mathbf{z}_0$.

In addition to $\mathcal{L}_{\text{rc}}$, the other objective $\mathcal{L}_{\text{kl}}$ is calculated for diffusion process through the approximated $\mathcal{D}_{\text{kl}}$ using $\mathcal{D}_{\text{prod}}$ and $\mathcal{D}_{\text{var}}$ where an approximation error tolerance η is applied. In practice, η controls the trade-off between approximate reliability

Algorithm 2: CMD Sampling Procedure.

Require : diffusion denoiser θ , input text $\mathbf{x}_0$, embed layer ϕ , variance schedule $\{\beta_t\}$

sample $\mathbf{z}_T = \mathbf{z}^x \oplus \mathbf{z}^y$ by $\mathbf{z}^x = g_\phi(\mathbf{x}_0)$ & $\mathbf{z}^y \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

for $t = T, \dots, 1$ **do**

 predict a denoiser output $\hat{\mathbf{z}}_0 = f_\theta(\mathbf{z}_t, t)$

for each token n in $\mathbf{z}^y$ **do**

 sample $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\epsilon = \mathbf{0}$

 sample $\rho_t^n \sim \text{Bernoulli}(\frac{t-1}{T})$

$\mu_\theta(\mathbf{z}_t^n, t) = \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})\mathbf{z}_t^n - \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\hat{\mathbf{z}}_0^n}{1 - \bar{\alpha}_t}$

$\mathbf{z}_{t-1}^n = \begin{cases} \mathbf{m}, & \text{if } \rho_t^n = 1 \\ \frac{1}{\sqrt{\alpha_t}}\mu_\theta(\mathbf{z}_t^n, t) + \beta_t\epsilon, & \text{if } \rho_t^n = 0 \end{cases}$

end

end

return $\mathbf{z}_0$ or $\mathbf{z}^{y_0}$ for $\mathbf{y}_0$

and training signal utilization. A larger η includes more KL terms but risks incorporating inaccurate approximation while a smaller η improves reliability but reduces effective training signals. Also, the mixture weights used in the KL approximation are computed from b_t and b_{t-1} according to (17), (18) and (31), which are used to calculate the transition distributions in (34) and (35) in the forward and reverse processes, respectively. These weights are instantiated at each diffusion step and directly used in (38) and (39). This connection ensures that the theoretical formulation is faithfully implemented in training. Notably, KL divergence between two Gaussian mixtures with two sets of means and covariances is involved. One set for q in (34) considers the mean at $\mathbf{m}$ with a small covariance $\sigma^2 \mathbf{I}$ as well as the mean and variance of the posterior similar to those in the original diffusion model [26]. The other set for p in (35) adopts the posterior means in a Gaussian mixture by using the predicted $\mathbf{m}$ and $\mathbf{z}_t^n$. Overall, $\mathcal{L}_{\text{tri}}$, $\mathcal{L}_{\text{kl}}$ and $\mathcal{L}_{\text{rc}}$ are used to update ϕ while $\mathcal{L}_{\text{kl}}$ and $\mathcal{L}_{\text{rc}}$ are used to update θ and $\mathbf{m}$. This process is repeated through sufficient number of iterations until convergence.

As shown in Algorithm 2, the sampling stage of the proposed CMD is performed by firstly concatenating the input embedding $\mathbf{z}^x$ from condition text $\mathbf{x}_0$ with the white noise $\mathbf{z}^y$ for target text $\mathbf{y}$ to form $\mathbf{z}_T$. During the denoising iterations, the denoiser f_θ provides a predicted clean sample $\hat{\mathbf{z}}_0$. This sample is then used to backtrack $\mathbf{z}_{t-1}^n$ from $\mathbf{z}_t^n$ at each step t based on the expected value of $p_\theta(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n)$ in (35) or specifically $q(\mathbf{z}_{t-1}^n | \mathbf{z}_t^n, \mathbf{z}_0^n)$ in (34) at either mask token $\mathbf{z}_{t-1}^n = \mathbf{m}$ with $\rho_t^n = 1$ or the denoised sample with $\rho_t^n = 0$ by referring (2) to derive

$$\mathbf{z}_{t-1}^n = \frac{1}{\sqrt{\alpha_t}} \mu_\theta(\mathbf{z}_t^n, t) + \beta_t \epsilon \quad (44)$$

where the mean vector is due to μ_4 in (27) in a form of

$$\mu_\theta(\mathbf{z}_t^n, t) = \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})\mathbf{z}_t^n - \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\hat{\mathbf{z}}_0^n}{1 - \bar{\alpha}_t}. \quad (45)$$

The mask indicator ρ_t^n is drawn from Bernoulli with a decreasing rate. The final sample $\mathbf{z}_0$ or $\mathbf{y}_0$ is then obtained. It is noted to connect b_t in (14), γ_t in Algorithm 1 and ρ_t^n in Algorithms 1 and 2. The mixture weight b_t represents the marginal probability of selecting the Dirac component while $\rho_t^n \sim \text{Bernoulli}(\gamma_t)$ is its stochastic indicator in binary value during training, and ρ_t^n in

TABLE I
EXAMPLES OF PARAPHRASING SENTENCES IN QQP

Condition sentence x_0	Target sentence y_0
Can aliens ever be real?	Do aliens exists?
How many stars are on the American flag?	How many stars in flag of United States?

TABLE II
EVALUATION OF PARAPHRASING RESULTS ON QQP OVER DIFFERENT METRICS USING RELATED METHODS AT DIFFERENT STEPS T

Method	Step	BLEU $\uparrow$	R-L $\uparrow$	BS $\uparrow$	Time(s) $\downarrow$
D3PM [9]	10	0.2245	0.5677	0.8281	–
DS [9]	2000	0.1884	0.5327	0.7965	–
DS-v2 [9]	2	0.2115	0.5651	0.8036	–
	10	0.2210	0.5732	0.8207	–
	2000	0.2441	0.5930	0.8393	–
DS (ours)	2000	0.1777	0.5298	0.7953	1245
DS-v2 (ours)	2	0.2089	0.5650	0.8039	2.86
	10	0.2180	0.5753	0.8206	7.68
	20	0.2182	0.5760	0.8232	13.84
	2000	0.2440	0.5959	0.8418	1074
CMD	2	0.2235	0.5785	0.8185	2.99
	10	0.2293	0.5890	0.8381	7.96
	20	0.2317	0.5838	0.8401	13.98
	2000	0.2460	0.5980	0.8501	1089

TABLE III
EVALUATION ON SEQ2SEQ (S2S) TASKS FOR OPEN DOMAIN DIALOGUE (D) AND TEXT SIMPLIFICATION (T) OVER ACCURACY, DIVERSITY AND HUMAN EVALUATION (HE). 10 STEPS ARE FIXED.

S2S	Method	BLEU $\uparrow$	R-L $\uparrow$	BS $\uparrow$	dist-1 $\uparrow$	div-4 $\uparrow$	HE $\uparrow$
D	D3PM	0.0190	0.1118	0.5305	0.9501	0.9961	1.36
	DS [3]	0.0139	0.1056	0.5131	0.9467	0.9971	1.52
	DS-v2	0.0199	0.1089	0.5283	0.9498	0.9983	1.69
	CMD	0.0220	0.1351	0.5432	0.9662	0.9989	1.88
T	D3PM	0.3719	0.5820	0.8098	0.9321	0.6710	1.19
	DS [3]	0.3622	0.5849	0.8126	0.9264	0.6604	1.45
	DS-v2	0.3711	0.5921	0.8201	0.9319	0.6801	1.65
	CMD	0.3805	0.6106	0.8385	0.9430	0.6993	1.84

sampling follows the same schedule. This connection ensures the consistency between training and inference in Algorithms 1 and 2, respectively.

IV. EXPERIMENTS

A. Experimental Settings

In the experiments, the diffusion denoiser f_θ followed the same network architecture as described in [9], which was a BERT base encoder with 12 layers and 12 heads per layer under a 768-dimensional latent space. The input variables $\mathbf{z}_t$ and t to denoiser f_θ were provided with separate embedding layers along with the positional encodings. The total model size was around 91 million parameters. The sequence length of a generation was fixed at 98, and the number of diffusion steps T was 2000. The noise variance $\beta_t = 1 - \alpha_t$ and the mask rate γ_t followed the schedule $\sqrt{t/T} + 10^{-4}$ [26] along different diffusion steps t . The computation time was measured on a personal computer equipped with a single RTX 4090 GPU. The hyperparameters were empirically specified for error tolerance as $\eta = 0.01$, triplet margin as $\xi = 10^{-4}$, and regularization factors as $\lambda_c = 0.2$, $\lambda_a = 0.35$ and $\lambda_r = 0.45$. Selecting the factors $\{\lambda_c, \lambda_a, \lambda_r\}$ was performed via a simple grid search on a held-out validation set

TABLE IV
EVALUATION ON GENERATION PERPLEXITY $\downarrow$ ON OWT VALIDATION DATA USING RELATED METHODS AT DIFFERENT STEPS T

Size (Step)	D3PM	DS	DS-v2	MD4	MDLM	CMD
S (2)	29.51	28.36	26.91	24.32	23.34	20.23
S (10)	29.02	26.86	25.53	23.92	22.83	19.22
S (20)	28.22	25.21	25.14	23.02	22.02	19.02
S (2000)	24.84	21.92	20.21	19.99	19.90	19.00
M (10)	20.35	18.50	17.02	16.64	16.32	14.21

to obtain the best prediction. The AdamW optimizer [39] was used with the weight decay factor 0.01 and the learning rate 10^{-4} under a linear schedule. Batch size was set as 256. The maximum length allowed for diffusion generation in a non-autoregressive text generation was 98.

The paraphrasing task using the Quora Question Pairs (QQP) dataset was utilized to evaluate the performance of using different methods for generating sentences with identical semantic content. QQP consisted of 400 K pairs of questions, each pair was labeled to indicate whether two questions conveyed the same meaning. Using these labels, the similar question pairs were selected as the training samples $\mathcal{D} = \{\mathbf{x}_0, \mathbf{y}_0\}$, as shown with the examples given in Table I. This dataset was used to evaluate that the model was learned to effectively paraphrase sentences. A variety of metrics were evaluated to ensure the accuracy and the diversity of text generation in the results. BLEU [40], ROUGE-L (R-L) [41] and BERTScore (BS) [42] provided the automated evaluation metrics for quantifying the similarity between generated text and reference text [3]. BLEU measures the precision of n -grams in the generated text. ROUGE-L focuses on the recall of long common subsequence (LCS) matches. BERTScore assesses the semantic similarity of the generated text. For these three metrics, the higher the better. In addition, the computation time (in seconds) for real-time generation over 2,500 test samples was measured to evaluate the system efficiency and practicality. By examining these metrics, we were able to thoroughly assess the system performance and the practical value.

Various text diffusion models including D3PM [2], DiffuSeq (DS) [3], DiffuSeq-v2 (DS-v2) [9], MD4 [14], MDLM [13], and the proposed CMD were investigated. The discrete diffusion model using D3PM was implemented by [2] and examined with the result by referring [9]. DiffuSeq [3] was implemented for conditional text generation where only continuous diffusion model based on Gaussian distribution was considered while DiffuSeq-v2 [9] was developed as a continuous diffusion consisting of Gaussian noise and mask noise where the learning objective was over-simplified without considering probabilistic modeling. MD4 [14] implies the masked discrete diffusion for discrete data where the denoising softmax network was parameterized by a probability vector and was optimized by a weighted integral of cross-entropy losses. MDLM [13] was exploited as a masked diffusion LM through an *interpolating discrete diffusion* between (estimated) clean and mask tokens where KL divergence in diffusion loss was treated or simplified as a weighted average of masked cross entropy losses under the tricks of zero masking probabilities and carry-over unmasking. The fast

TABLE V
ABLATION RESULTS ON KL LOSS AND TRIPLET LOSS IN
CONTINUOUS-DISCRETE DENOISING PROCESS. PERFORMANCE DROP RATES
(%) BY $\mathcal{L}_{KL}$ OR/AND $\mathcal{L}_{TRI}$ ARE SHOWN IN THE BRACKETS.

$\mathcal{L}_{KL}$	$\mathcal{L}_{TRI}$	Steps	BLEU $\uparrow$	R-L $\uparrow$	BS $\uparrow$
×	×	2	0.2141 (-4.39)	0.5686 (-1.74)	0.8088 (-1.20)
		10	0.2211 (-3.71)	0.5807 (-1.43)	0.8301 (-0.96)
		20	0.2231 (-3.86)	0.5825 (-1.31)	0.8330 (-0.85)
		2000	0.2343 (-4.99)	0.5884 (-1.63)	0.8382 (-1.42)
✓	×	2	0.2194 (-1.87)	0.5729 (-0.98)	0.8120 (-0.80)
		10	0.2241 (-2.32)	0.5809 (-1.39)	0.8319 (-0.75)
		20	0.2263 (-2.39)	0.5825 (-1.31)	0.8352 (-0.59)
		2000	0.2369 (-3.84)	0.5921 (-1.00)	0.8413 (-1.05)
×	✓	2	0.2195 (-1.82)	0.5747 (-0.66)	0.8142 (-0.53)
		10	0.2253 (-1.78)	0.5840 (-0.86)	0.8353 (-0.34)
		20	0.2281 (-1.58)	0.5864 (-0.63)	0.8383 (-0.22)
		2000	0.2454 (-0.24)	0.5948 (-0.54)	0.8475 (-0.31)
✓	✓	2	0.2235	0.5785	0.8185
		10	0.2293	0.5890	0.8381
		20	0.2317	0.5901	0.8401
		2000	0.2460	0.5980	0.8501

TABLE VI
COMPARISON OF THE GENERATED SEQUENCES OF AN EXAMPLE IN THE
FORWARD PROCESS GIVEN BY THE CONDITION TEXT “WHAT CONTRIBUTION
HAVE INDIAN MATHEMATICIANS MADE TO MATHEMATICS IN GENERAL?” BY
USING DS-V2 AND CMD

Step	Forward generation by DS-v2
2	[CLS] What contribution have the Indian mathematicians made to mathematics ? [SEP]
100	[CLS] What contribution have [M] [M] [M] [M] in mathematicians ? [SEP]
200	[CLS] What contribution have [M] [M] [M] [M] in mathematicians ? [SEP]
300	[CLS] What [M] have [M] [M] [M] [M] in mathematicians ? [SEP]
400	[CLS] What [M] have [M] [M] [M] [M] mathematicians ? [SEP]
500	[CLS] What [M] have [M] [M] [M] [M] mathematicians ? [SEP]
600	[CLS] What [M] have [M] [M] [M] [M] mathematicians ? [SEP]
...	...
2000	[CLS] What [M] have [M] [M] [M] [M] [M] [M] ? [PAD]
Step	Forward generation by CMD
2	[CLS] What contributions to mathematics have been made by Indian mathematicians ? [SEP]
100	[CLS] What contributions to [M] [M] been made [M] Indian mathematicians ? [M]
200	[M] What [M] to [M] [M] been made [M] [M] [M] [M] [M]
300	[M] What [M] to [M] [M] been made [M] [M] [M] [M] [M]
400	[M] What [M] to [M] [M] [M] [M] [M] [M] [M] [M] [M]
500	[M] [M] [M] or [M] [M] [M] [M] [M] [M] [M] [M] [M]
600	[M] [M] [M] [M] [M] [M] [M] [M] [M] [M] [M] [M]
...	...
2000	[M] [M] [M] [M] [M] [M] [M] [M] [M] [M] [M] [M]

sampler based on ordinary differential equation (ODE) solver was used in different methods [5]. There are twofold advantages of using CMD relative to the previous methods, which need to be investigated. First, this study develops a Gaussian-Dirac denoising process for continuous diffusion, which carries out a probabilistic mixture diffusion model without simplification of KL divergence in alignment loss for masked diffusion. Second, the contrastive metric learning was incorporated to reflect the easy-first property as a strategy to text generation in masked diffusion.

In addition to paraphrasing, Seq2Seq tasks [3] on the open domain dialogue for generating informative responses given a dialogue as well as the text simplification for simplifying a

complex text into words with simplified grammar were evaluated. The intra-diversity using distinct unigram (dist-1) and the sentence-level diversity using diverse 4-gram (div-4) [43] were measured. Human evaluation (HE) ($\uparrow$) on the generated sentences was conducted by rating 0 (poor: incorrect or incoherent), 1 (acceptable: minor issues but understandable) or 2 (good: fluent and semantically consistent) as a 3-point scale according to the overall judgment of relevance and informativeness. There were five anonymous annotators who blindly and independently judged 100 randomly shuffled and sampled test sentences from individual methods. Inter-annotator agreement (IAA) was measured by using Fleiss’ kappa, yielding a score 0.682 indicating the substantial agreement. This study further evaluated the generation perplexity on OpenWebText (OWT) validation data [13], [14], 2% of original training data, by using different diffusion-based LMs, trained by referring to the settings in [44] in GPT-2 small (S) and medium (M) sizes.

B. Experimental Results

First of all, Table II demonstrates the comparison of D3PM [2], DS [3], DS-v2 [9] and CMD for text paraphrasing on the QQP dataset where the metrics of BLEU, ROUGE-L, BERTScore and inference time over 2,500 test samples are evaluated. For DS and DS-v2, the results from [9] and our results are both reported. Number of diffusion steps T is shown. The diffusion probabilistic model (DPM) solver++ [7] was employed in the accelerated sampling on continuous diffusion. The rapid samples with 10 steps were sufficient to ensure high-quality result. Experimental results illustrate that CMD outperforms DS-v2 on both BLEU and ROUGE-L scores. For BERTScore, CMD maintains a clear advantage over DS-v2 in the cases with fewer sampling steps, while performing comparable in cases with a larger number of sampling steps. The performance improvements in BLEU and ROUGE-L suggest that CMD generates the paraphrases with better fluency and closer alignment to the reference text. This is likely due to the probabilistic mixture diffusion in Seq2Seq task as well as the easy-first embedding structure by contrastive learning. However, it is found that since the contrastive embedding primarily impacts the generation performance during the early sampling stages, the advantages of CMD may be decayed as the number of diffusion steps increases. Overall, the easy-first strategy in CMD appears to provide a robust structure for text diffusion, which leads to an effective paraphrasing where higher BLEU, ROUGE-L and BERTScore are obtained with fewer diffusion steps relative to DS and DS-v2. While the proposed CMD provides efficiency in diffusion process, it is observed that when the sampling process is extended, the rigid structure may be changed at some expense of reflecting the semantic similarity in learning representation. Notably, the computation cost of using CMD is found to be comparable with that using DS and DS-v2. This CMD performs better than the discrete diffusion based on D3PM.

Table III extends the comparison over D3PM, DS, DS-v2 and CMD on the tasks for open domain dialogue and text simplification where the additional metrics using dist-1, div-4 and HE are measured. The samples with 10 steps are evaluated.

TABLE VII

[illegible]

CMD consistently obtains higher scores in objective quality and diversity, and subjective human evaluation on the generated text than DS-v2. In this comparison, CMD achieves the best results in terms of different metrics.

Table IV reports the generation perplexity on OWT validation set under different diffusion steps where various diffusion-based LMs (D3PM [2], DS [3], DS-v2 [9], MD4 [14], MDLM [13], and CMD) in small and medium sizes are consistently trained by 800 K updating iterations. Our implementation of different methods follows the same settings as in [14], [44]. Basically, different methods reduce the test perplexity by increasing diffusion steps. In case of 10 steps with GPT-2 small (S) size, it is found that CMD considerably reduces the perplexity by 24.7%, 19.7%, 15.8% relative to DS-v2, MD4 and MDLM respectively. In the comparison, the lowest perplexity 14.21 is obtained by using CMD in GPT-2 medium (M) size with 10 steps. The possible reasons why CMD surpasses the other methods are due to the enhancement of easy-first property within a few steps of diffusion process and the precise calculation of diffusion loss with the mixture densities for continuous-discrete noises. Improvement of CMD is reduced with 2000 steps. A recent work on remasking strategy [45] is helpful to recover wrong generation during long diffusion process. The easy-first property in CMD is useful for time-constrained generation.

C. Experimental Analyses

Relative to the recent DS and DS-v2, there are two innovations, which are the contrastive learning for the embedding space and the approximation to KL divergence between two mixture distributions. This paper conducted the ablation studies

on the performance of using different learning objectives by training different number of denoising steps. These analyses would like to provide a profound understanding of the strengths and limitations of the proposed method as well as the insights into its potential improvements. Table V reports the ablation results on merging the stepwise KL loss $\mathcal{L}_{kl}$ for alignment between forward and reverse processes and the triplet loss $\mathcal{L}_{tri}$ for fulfillment of easy-first generation steps. When both KL loss and triplet loss are disregarded, the resulting variant turns out as a solution, which minimizes the reconstruction loss $\mathcal{L}_{rc}$. Such a solution is similar to DS-v2, and its performance drops significantly across all metrics. Adding only KL loss slightly improves the metrics but not as effectively as including KL and triplet losses. Including only triplet loss results in better improvements than including KL loss alone. The triplet loss introduces additional computational cost due to the pairwise comparisons between positive and negative samples. In practice, we mitigate this cost by sampling subsets of the vocabulary, reducing the complexity from quadratic in vocabulary size to a manageable level. While this loss increases the training cost compared to DS-v2, the inference efficiency remains comparable, and the improved generation quality justifies the trade-off. In this set of experiments, the highest scores are obtained due to the combined benefits from both learning objectives. This ablation study highlights the importance of minimizing individual losses to achieve the best results.

Next, Tables VI and VII illustrate the generated sequences at different steps in the forward and denoising processes using DS-v2 and CMD given by the examples of condition sequences “What contribution have Indian mathematicians made to mathematics in general?” and “What were the major effects of

the Cambodia earthquake, and how do these effects compare to the Banda Sea earthquake in 1938?”, respectively. Here, [M], [SEP], [CLS], [PAD] denote the special tokens to imply mask, separation, classification and padding, respectively. In the first example for noising process, DS-v2 converges very slowly while CMD generates a sequence of masks rapidly. The common words “what” and “to” turn to mask tokens lately in CMD. In the second example for denoising process, DS-v2 generates key terms like “Cambodia” and “effects” in the earlier steps by following a kind of balanced and gradual unmasking strategy where a smoother refinement of sentence generation is implemented. The formation of a clear structure of a complex sentence has been slowed down. On the other hand, CMD fulfills an easy-first strategy which results in generating the common words like “what” and “effects” earlier while generating more specific terms such as “Banda Sea” until later steps. This strategy helps establish a basic sentence structure before incorporating specific information. The main difference relative to DS-v2 is that CMD lays down a simple foundation faster and earlier, which can be advantageous when dealing with more detailed and complicated inputs. Under the same number of diffusion steps, CMD demonstrates faster than DS-v2 to complete the masking and unmasking processes.

V. CONCLUSION

This study explored the integration of continuous and discrete diffusion models, emphasizing their potential to enhance text generation capabilities. This research highlighted the effectiveness of easy-first generation strategy, coupled with contrastive learning, to improve the efficiency and quality of the generated text. By leveraging the strengths of both continuous and discrete models, this paper presented a hybrid approach to address the inherent limitations of existing generative models. Our findings demonstrated that continuous diffusion models, when supplemented with discrete state mechanism, offered superior sampling efficiency and flexibility. This approach allowed for smoother transitions and more nuanced text generation by interpolating between word embeddings in diffusion steps. The introduction of contrastive learning further refined the model’s ability to prioritize embeddings properly, ensuring that higher-frequency and simpler words were generated earlier. This structured embedding space aligned with the easy-first generation, facilitating efficient and high-quality text output. The implications of easy-first principle extended the text generation beyond merely word-frequency level. The applications for text paraphrasing and other tasks could be adapted to prioritize various attributes such as empathy or sentiment, enabling contextually and emotionally appropriate generation for empathetic dialogue. By focusing on generating the content that aligns with the desired attributes, this approach provides the nuanced control over a structural generation. The hybrid continuous-discrete diffusion model, enhanced with contrastive learning, not only improves the likelihood of generating high-quality text but also offers a flexible and efficient framework for text generation. These advancements pave the way to future exploration and innovation, including attribute-based text generation, thereby

broadening the scope and applicability in content creation. This study advances and evolves the generation process to build a sophisticated and smoothed diffusion model.

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Jen-Tzung Chien (Senior Member, IEEE) is currently the Lifetime Chair Professor with National Yang Ming Chiao Tung University, Taiwan. He has authored more than 300 peer-reviewed articles in machine learning, natural language processing and computer vision, and three books including *Bayesian Speech and Language Processing*, Cambridge University Press, 2015, *Source Separation and Machine Learning*, Academic Press, 2018, and *Machine Learning for Speaker Recognition*, Cambridge University Press, 2020. He was a Tutorial Speaker of AAAI, IJCAI, ACL, MM, KDD, ICASSP, CIKM, WSDM, COLING, and Interspeech. He received the Best Paper Award in *IEEE Workshop on Automatic Speech Recognition and Understanding* in 2011, and *IEEE International Workshop on Machine Learning for Signal Processing* in 2023. He serves as an Elected Member of the *IEEE Machine Learning for Signal Processing Technical Committee* and the *IEEE Speech and Language Processing Technical Committee*, and the Associate Editor of the *IEEE Transactions on Audio, Speech and Language Processing*.



Chih-Chun Chen received the BS degree in mathematics from National Tsing Hua University, in 2022, and the MS degree in electrical and computer engineering from National Yang Ming Chiao Tung University, in 2024. His research interest includes statistical learning for large language models.